

Dahua Feng

Website / GitHub

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Education

M.S. in Electronics Engineering, KAIST

Department of Electronics Engineering

B.S. in Information and Computing Sciences, Peking University

School of Electronics Engineering and Computer Science

Daejeon, South Korea

Sep. 2025 - Jul. 2027 (expected)

Beijing, China

Sep. 2020 - Jul. 2024

- **GPA:** 3.700/4.000 (87.4/100)
- **Thesis:** *A Simulator Design for the Storage of Mobile Devices* (supervised by Prof. Jie Zhang)

Research Experience

SEPT Lab, University of Southern California

Research Assistant

Project: CPU-GPU communication overhead in confidential computing environments

Jun. 2025–Nov. 2025

Advisor: Prof. Mengyuan Li

- Investigated CPU-GPU communication overheads introduced by confidential computing (CC) for LLM inference and fine-tuning in GPU TEEs; identified the CC authentication protocol as a serialization bottleneck; aimed to reduce the overhead of CC via protocol redesign.
- Implemented a block-indexed parallel authentication scheme and a lazy-decryption mechanism to avoid unnecessary decrypting/re-encrypting of unused data.
- Conducted a literature survey and theoretical analysis, implemented the end-to-end system for the scheme, and evaluated its performance.

XSEL Lab, University of Virginia

Research Assistant

Project: Apple Silicon for machine learning

Aug. 2024–Dec. 2024

Advisor: Prof. Felix Lin

- Focused on evaluating ML training performance on Apple Silicon to support affordable, democratized ML training; aimed to tap into large memory of Apple Silicon and lower the barrier to entry for ML training; profiled and analyzed performance gaps with NVIDIA GPUs, emphasizing LLM workloads.
- Executed end-to-end training experiments on state-of-the-art generative models and micro-benchmarked BLAS kernels across multiple hardware platforms.
- Consolidated and analyzed benchmark data, producing insights and actionable recommendations for practitioners.

Picasso Lab, University of California, Santa Barbara

Research Intern

Project: CXL-based memory-disaggregated system for DLRM

Jul. 2023–Dec. 2023

Advisor: Prof. Yufei Ding

- Designed and evaluated a CXL-backed memory disaggregation architecture to accelerate DLRM training by sharding embedding tables across GPUs and CXL memory expanders.
- Formally analyzed embedding-table placement and proposed load-balanced memory-allocation algorithms to minimize remote accesses and contention across GPUs.
- Surveyed ML-based and heuristic sharding methods (including RL) and identified practical improvements to reduce communication overhead and improve utilization.

Project: DLRM embedding lookup with NVSHMEM communication

- Integrated NVSHMEM as an inter-GPU communication layer to support asymmetric embedding-table placement and reduce remote lookup latency.
- Collected realistic kernel-level profiling data via CUDA kernels to quantify the impact of placement and batching strategies.
- Built a latency-prediction model for batched embedding lookups and used it to guide placement and batching decisions for improved throughput.

Teaching Experience

Teaching Assistant of Computer Architectures

Fall 2023

School of Electronic Engineering and Computer Science, Peking University

- Assisted Prof. Jie Zhang in designing and refining course projects; led Q&A sessions; prepared and administered quizzes; contributed to final exam design and grading.

Skills

Programming Languages & Software: C, C++, Python, CUDA C

Python Packages: torch, scikit-learn

Languages: Mandarin (native), English (proficient), Korean (intermediate)